

★★★★★ Tier 1 (Must Master)

These account for **70-80% of stack interview questions**.

Pattern 1 — Monotonic Stack (Most Important) ★★★★★

Concepts

Know when to use it:

- Next Greater Element
- Next Smaller Element
- Previous Greater Element
- Previous Smaller Element
- First greater/smaller on left/right
- Increasing stack
- Decreasing stack
- Circular array variation

Recognize by clues:

- "Nearest"
- "First greater"
- "First smaller"
- $O(n^2)$ brute force can become $O(n)$

Interview Questions

Easy

- Next Greater Element I (LC 496) DONE
- Next Greater Element II (LC 503) DONE
- Daily Temperatures (LC 739) DONE

Medium

- Final Prices With Special Discount (LC 1475) DONE
- Online Stock Span (LC 901) DONE
- Remove K Digits (uses monotonic stack) PATTERN 3

Hard

- Trapping Rain Water (stack solution) DONE
- Largest Rectangle in Histogram PATTERN 2

Pattern 2 — Histogram / Area Pattern ★★★★★

Built completely on Monotonic Stack.

Concepts

Know:

- Previous Smaller
- Next Smaller
- Width calculation
- Rectangle expansion
- Area = height × width

Formula

```
width = nextSmaller - prevSmaller - 1
```

Interview Questions

Must Do

- Largest Rectangle in Histogram (LC 84) DONE
- Maximal Rectangle (LC 85) DONE

Good Practice

- Sum of Subarray Minimums PATTERN 4

Pattern 3 — Stack + Greedy ★★★★★

Very common.

Concepts

Know:

- Remove bad choices
- Keep best answer
- Lexicographically smallest/largest
- Pop while current element is better

Interview Questions

Must Do

- Remove K Digits (LC 402) DONE
- Remove Duplicate Letters (LC 316) DONE
- Smallest Subsequence of Distinct Characters (LC 1081) DONE

Practice

- Create Maximum Number HARD TBD
- Remove All Adjacent Duplicates II DONE

Pattern 4 — Contribution / Subarray Stack ★★★★★

Frequently asked in good companies.

Concepts

Instead of checking every subarray,

Calculate each element's contribution.

Need:

- Previous Smaller
- Next Smaller
- Previous Greater
- Next Greater

Contribution formula

Contribution =
Left Choices × Right Choices × value

Interview Questions

Must Do

- Sum of Subarray Minimums (LC 907) DONE
- Sum of Subarray Ranges (LC 2104) DONE

Practice

- Total Strength of Wizards (Hard)

★★★★★ Tier 2 (Very Important)

Pattern 5 — Parentheses / Bracket Validation ★★★★★

Classic stack questions.

Concepts

Know:

- Matching brackets
- Balanced parentheses
- Nested expressions
- Store indices
- Store characters

Interview Questions

Easy

- Valid Parentheses (LC 20)

Medium

- Minimum Remove to Make Valid Parentheses DONE
- Remove Outermost Parentheses DONE
- Score of Parentheses TBD
- Decode String DONE

Hard

- Longest Valid Parentheses TBD
-

Pattern 6 — Stack Design & Simulation ★★★★★

Tests implementation skills.

Concepts

Know:

- Custom stack
- Min Stack
- Max Stack
- Two stacks
- Queue using stacks
- Stack using queues

Interview Questions

Must Do

- Min Stack (LC 155)
- Implement Queue using Stacks (LC 232)
- Implement Stack using Queues (LC 225)

Good Practice

- Design Browser History
- Dinner Plate Stacks

Pattern 7 — Expression Evaluation ★★★★★

Less frequent but standard.

Concepts

Know

- Prefix
- Infix
- Postfix
- Operator precedence
- Operand stack
- Operator stack

Interview Questions

Must Do

- Evaluate Reverse Polish Notation (LC 150)

Practice

- Basic Calculator I
- Basic Calculator II
- Basic Calculator III

★★★★ Tier 3 (Good to Know)

Pattern 8 — Stack + String Processing ★★★★★

Concepts

Know

- Remove duplicates
- Reverse processing
- Backtracking using stack
- Character manipulation

Interview Questions

- Backspace String Compare
- Remove All Adjacent Duplicates
- Make The String Great
- Simplify Path

Pattern 9 — Stock Span Pattern ★★★★★

Technically a direct application of Monotonic Stack.

Concepts

Know

- Previous Greater Element
- Span calculation

Formula

span = currentIndex - previousGreaterIndex

Interview Questions

- Online Stock Span (LC 901)
- Stock Span Problem (GFG)

Pattern 10 — Stack for DFS / Iterative Traversal ★ ★ ★

Mostly useful in Graphs and Trees rather than Stack interviews.

Concepts

Know

- Replace recursion with stack
- Iterative DFS
- Preorder
- Inorder
- Postorder (two stacks / one stack)

Interview Questions

Trees

- Binary Tree Preorder Traversal
- Binary Tree Inorder Traversal
- Binary Tree Postorder Traversal

Graphs

- DFS of Graph
- Number of Islands (iterative)

 Interview Preparation Order

If you have limited time, follow this sequence:

PriorityPatternInterview Importance ★ ★ ★ ★ ★ Monotonic StackMust Master ★ ★ ★ ★ ★ Histogram / AreaMust Master ★ ★ ★ ★ ★ Stack + GreedyMust Master ★ ★ ★ ★ ★ Contribution / SubarrayMust Master ★ ★ ★ ★ ★ ParenthesesVery Important ★ ★ ★ ★ ★ Stack DesignVery Important ★ ★ ★ ★ ★ Expression EvaluationImportant ★ ★ ★ ★ ★ Stack + StringGood to Know ★ ★ ★ ★ ★ Stock SpanGood Practice ★ ★ ★ ★ ★ DFS using StackMostly for Graphs

 Recommended LeetCode Practice List (18 Problems)

Must Do (Core 12)

1. Valid Parentheses (20)
2. Min Stack (155)
3. Next Greater Element I (496)
4. Next Greater Element II (503)
5. Daily Temperatures (739)
6. Online Stock Span (901)
7. Largest Rectangle in Histogram (84)
8. Maximal Rectangle (85)
9. Remove K Digits (402)
- 10.Remove Duplicate Letters (316)
- 11.Sum of Subarray Minimums (907)
- 12.Evaluate Reverse Polish Notation (150)

Strong Practice (6)

1. Trapping Rain Water (42) – Stack solution
2. Sum of Subarray Ranges (2104)
3. Decode String (394)
4. Basic Calculator II (227)
5. Longest Valid Parentheses (32)
6. Simplify Path (71)

Mastering these patterns and problems will prepare you for the vast majority of stack-based interview questions in software engineering interviews.

